Table of Fourier Transform Pairs

$x(t) = \mathcal{F}^{-1}\{X(f)\} = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}\,df$	$\longleftrightarrow^{\mathcal{F}}$	$X(f) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\,dt$	
transform	$x(t)$	$X(f)$	
complex conjugation	$x^*(t)$	$X^*(-f)$	
time shifting	$x(t-t_0)$	$e^{-j2\pi ft_0}X(f)$	
	$e^{j2\pi f_0 t}x(t)$	$X(f-f_0)$	frequency shifting
time scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{f}{a}\right)$	
reversal	$x(-t)$	$X(-f)$	
duality (symmetry)	$X(-t)$	$x(f)$	
linearity	$ax_1(t) + bx_2(t)$	$aX_1(f) + bX_2(f)$	
multiplication	$x_1(t)x_2(t)$	$X_1(f) * X_2(f) = \int_{-\infty}^{\infty} X_1(\zeta)X_2(f-\zeta)\,d\zeta$	
convolution	$x_1(t) * x_2(t)$	$X_1(f)X_2(f)$	
	$= \int_{-\infty}^{\infty} x_1(\lambda)x_2(t-\lambda)\,d\lambda$		
delta function	$\delta(t)$	1	
shifted delta function	$\delta(t-t_0)$	$e^{-j2\pi ft_0}$	exponential decay
unit doublet	$\dot{\delta}(t)$	$j2\pi f$	
unit step	$u(t)$	$\frac{1}{2}\delta(f) + \frac{1}{j2\pi f}$	
signum	$sgn(t) = \begin{cases} 1 & t > 0 \\ -1 & t < 0 \end{cases}$	$\frac{1}{j\pi f}$	
exponential decay	$e^{-at}u(t)$	$\frac{1}{a+j2\pi f}$	
two-sided exp. decay	$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2+4\pi^2 f^2}$	
gaussian	$e^{-a^2t^2}$	$\frac{\sqrt{\pi}}{a}e^{-\frac{4\pi^2 f^2}{a^2}}$	
sine	$\sin(2\pi f_0 t)$	$\frac{1}{j2}\left[\delta(f-f_0) - \delta(f+f_0)\right]$	
cosine	$\cos(2\pi f_0 t)$	$\frac{1}{2}\left[\delta(f-f_0) + \delta(f+f_0)\right]$	
sinc	$F\mathrm{sinc}(Ft)$	$\Pi_F(f) = \mathrm{rect}\left(\frac{f}{F}\right)$	
gate function	$\Pi_T(t) = \mathrm{rect}\left(\frac{t}{T}\right)$	$T\mathrm{sinc}(Tf) = \frac{T\sin(\pi Tf)}{\pi Tf}$	
triangular function	$\Lambda_T(t) = \begin{cases} \left(1 - \frac{\lvert t\rvert}{T}\right) & \lvert t\rvert < T \\ 0 & \text{else} \end{cases}$	$T\mathrm{sinc}^2(Tf)$	
	$tx(t)$	$\left(\frac{j}{2\pi}\right)\frac{d}{df}X(f)$	frequency differentiation
	$t^n x(t)$	$\left(\frac{j}{2\pi}\right)^n \frac{d^n}{df^n}X(f)$	freq. n-th differentiation
time differentiation	$\frac{d}{dt}x(t)$	$j2\pi fX(f)$	
time n-th differentiation	$\frac{d^n}{dt^n}x(t)$	$(j2\pi f)^n X(f)$	
integration	$\int_{-\infty}^{t} x(t)$	$\frac{1}{j2\pi f}X(f) + \frac{1}{2}X(0)\delta(f)$	
modulation	$x(t)\cos(2\pi f_0 t)$	$\frac{1}{2}\left[X(f-f_0) + X(f+f_0)\right]$	